

Final Project – Natural Language Processing

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1. General Policy

The project must be completed in **pairs** (Any exception requires instructor approval).

Restrictions:

- Using Kaggle (datasets, notebooks, competitions, or code) is **not allowed**.
- Overused tutorial datasets (e.g., default IMDB demos without novelty) are discouraged unless a clearly original research angle is presented.

Originality in dataset selection, research question, and modeling strategy will be positively evaluated.

2. Project Tracks

Students must choose one of the following tracks:

Track 1 Educational Advanced NLP Project

Independent study of an advanced NLP method not covered in class.

- **Mandatory:** Clear theoretical explanation, mathematical/algorithmic description, code implementation, empirical demonstration on real data, quantitative evaluation, and critical discussion.
- **Warning:** A purely theoretical presentation is not acceptable. Slides without experiments will receive low credit.

Track 2 Research Paper Reproduction

Students select a peer-reviewed NLP paper.

- **Mandatory:** Reproduce at least one core experiment, implement key components independently, and compare results with the original paper.
- **Requirement:** Attempt a *meaningful improvement* (e.g., hyperparameter tuning, alternative feature representation, different dataset, modified architecture, error reduction analysis).
- **Warning:** Mere summary of the article is not acceptable. Exact reproduction without improvement is insufficient.

Track 3 Data-Driven NLP Project

Full NLP pipeline required.

- **Mandatory:** Data collection, preprocessing, feature engineering, at least two modeling approaches, evaluation, interpretation, and error analysis.
- **Encouraged:** Hebrew NLP, Domain-specific corpora, self-collected data, under-analyzed sources.

3. Technical Requirements (Mandatory for All)

Each project must include:

- Python implementation.
- Google Colab link OR GitHub repository.
- Email submission including ZIP archive.
- Reproducible code with clear train/test split.
- At least **two evaluation metrics**.
- **Error analysis:** Minimum 10 analyzed examples.
- At least 2 meaningful visualizations.

Students must be able to explain every part of their code.

4. Submission Format

Required Items:

- **Link:** Google Colab (fully executable) or GitHub repository.
- **Email Submission:** ZIP archive containing the final report (PDF, max 5 pages), slides, dataset description, and AI Usage Disclosure section.

Submission checklist (must be included in the ZIP):

- Report (PDF) – max 5 pages
- Slides (PDF or PPTX)
- Code (Colab notebook export or repository snapshot)
- AI_USAGE.md – full disclosure
- REFLECTION.md – personal reflection
- ETHICS.md – ethical statement
- DATA.md – dataset source, license, and reproduction steps

Submission method: email + link (Colab/GitHub). The exact email address and deadlines will be announced on Moodle.

Projects must be fully reproducible.

5. AI Usage Policy (Mandatory Disclosure)

AI tools may be used only as support tools. Every project must include a section named "**AI Usage Disclosure**". This disclosure must appear in **ALL** of the following places:

- **Report:** dedicated section
- **Notebook/Colab:** first markdown cell titled "AI Usage Disclosure"
- **Slides:** final slide titled "AI Usage Disclosure"
- **Repository/ZIP:** AI_USAGE.md file

Students must explicitly state:

- Which AI tools were used.

- For what purpose and to what extent.
- What was rewritten or modified manually.
- What was independently verified.

Required Format: Tool / Task / Extent / Human edits / Verification

Undisclosed AI usage constitutes academic misconduct. Students must understand and explain all submitted code.

6. Reflection & Technical Ownership (Mandatory)

Each report must include a **Personal Reflection and Technical Ownership** section (Min. one full page).

Students must describe:

- What they personally implemented.
- What failed and how they fixed it.
- Technical challenges encountered.
- What they learned and what surprised them.
- What they would improve.
- **Failure log:** 2–3 concrete things that failed and how they fixed them.
- **Team contribution log:** What each teammate did; short work log.

This section has significant grading weight.

7. Ethical Considerations (Mandatory)

Each project must include an **Ethical and Responsible NLP Statement** addressing:

- Data source transparency.
- Privacy considerations.

- Bias in data and model limitations.
- Potential harmful applications.
- Data licensing/copyright and permitted use.
- Consent/public availability (especially for social media data).

Projects that ignore ethical implications will receive lower evaluation.

8. Oral Technical Mini-Defense

After the presentation, each team will undergo a 5-minute technical Q&A.

Possible questions:

- Why did you choose this metric?
- What happens if hyperparameter X changes?
- Why did the model fail in example Y?
- What are the limitations of your approach?

Performance in the mini-defense affects the Reflection score. This ensures authentic understanding and technical ownership.

9. Timeline

- **Week 4:** Project proposal (max 2 pages). Includes track selection, research question, dataset, methods, and challenges. *Instructor approval required.*
- **Week 5–6:** Feedback and scope refinement.
- **Final week of the semester:** Presentation (20–30 minutes, 20–40 slides). Exact dates on Moodle.
- **Final Submission:** Up to 4 weeks after presentation. No late submissions without approval.
- **Calendar dates:** Exact deadlines will be announced on Moodle.

10. Grading Rubric (50% of Course Grade)

This rubric sums to 100% and is applied to the project grade (which constitutes 50% of the course grade).

Component	Weight
Proposal quality	10%
Technical implementation	20%
Experimental rigor	15%
Improvement / innovation	10%
Data originality	5%
Written report	10%
Presentation	10%
AI transparency	5%
Reflection + mini-defense	15%
Total	100%

11. Academic Integrity

The following are **prohibited**:

- Full AI-generated projects.
- Undocumented AI assistance.
- Copy-paste tutorials.
- Any use of Kaggle (datasets, notebooks, competitions, or code).
- Using transformer pipelines without analysis.

12. Bonus (Up to +5%)

Awarded for highly original datasets, Hebrew NLP contribution, cross-method comparisons, strong negative result analysis, or advanced error analysis.

Important Note: The final grade cannot exceed 100%, even if bonuses are awarded.